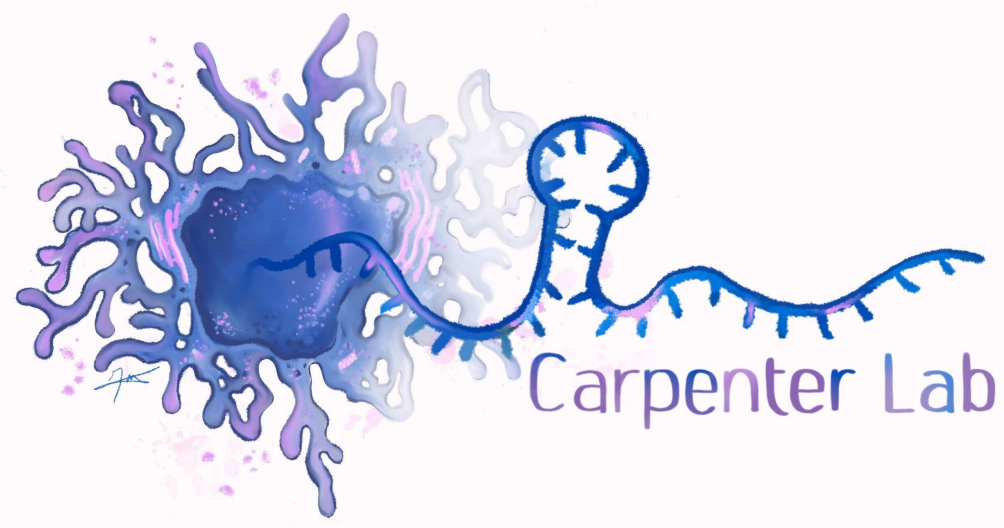




Predicting Ribo-seq signal from sequence and RNA-seq

Eric Malekos¹, Susan Carpenter²

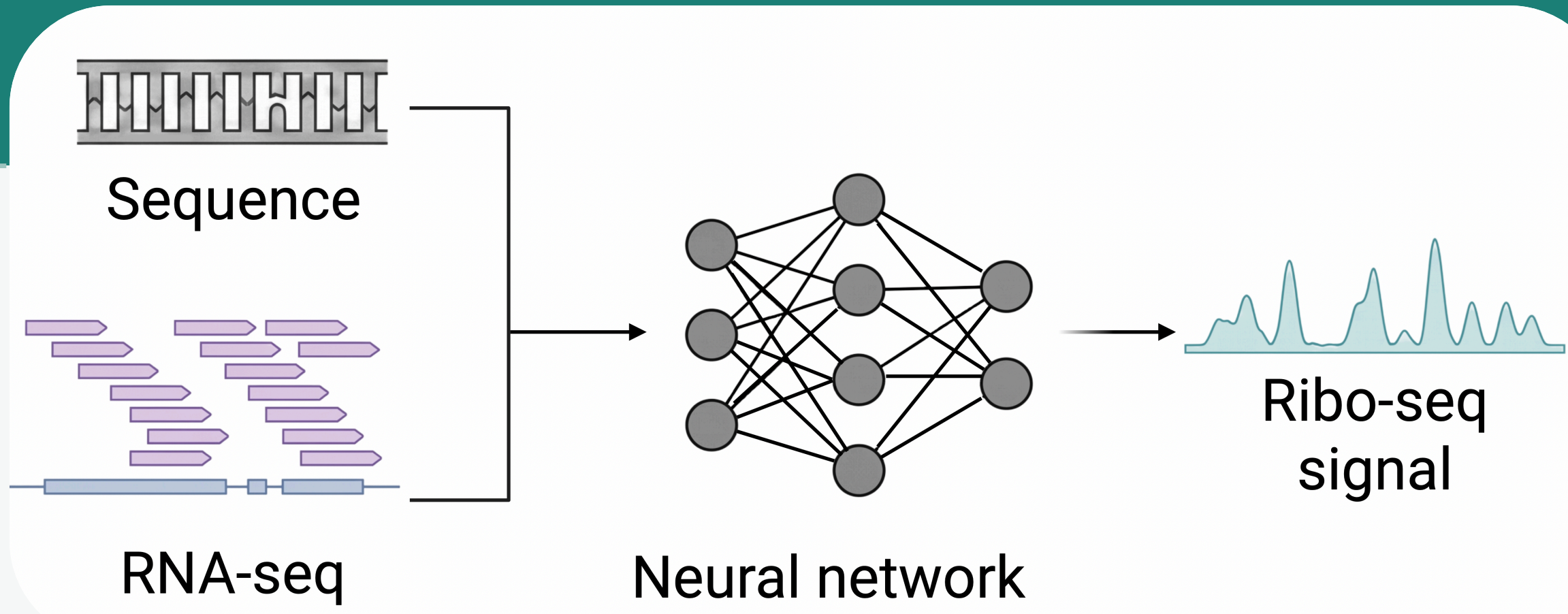
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PREDICTING RIBOSOME PROFILING SIGNAL WITHOUT RIBOSOME PROFILING

METHODS

- Predict per-nucleotide P-site density from sequence, a candidate-ORF track and RNA-seq coverage
- Feed the predicted profile into RiboCode to call ORFs with no Ribo-seq for the query
- Evaluate on held-out tissues and independent studies

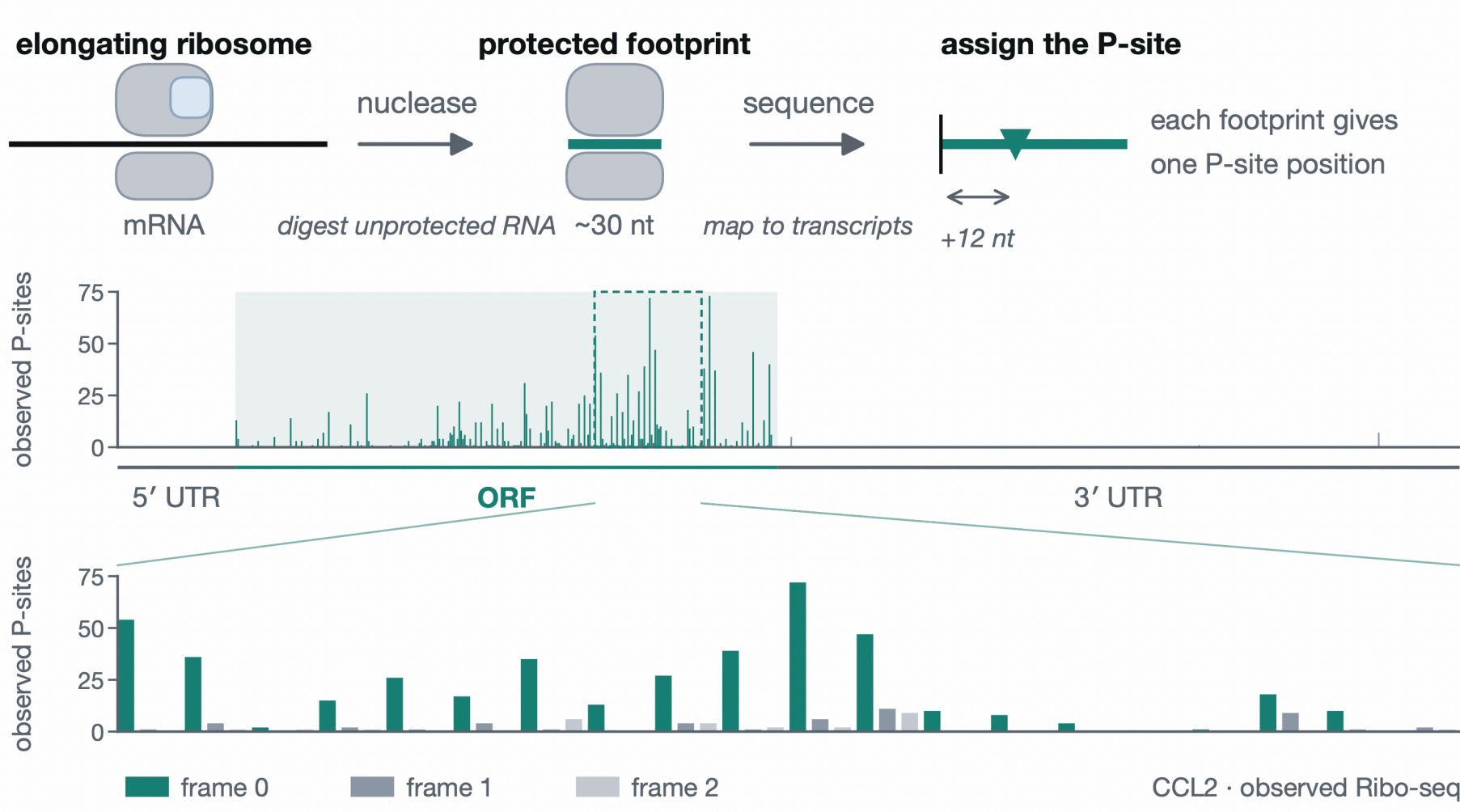


AIMS

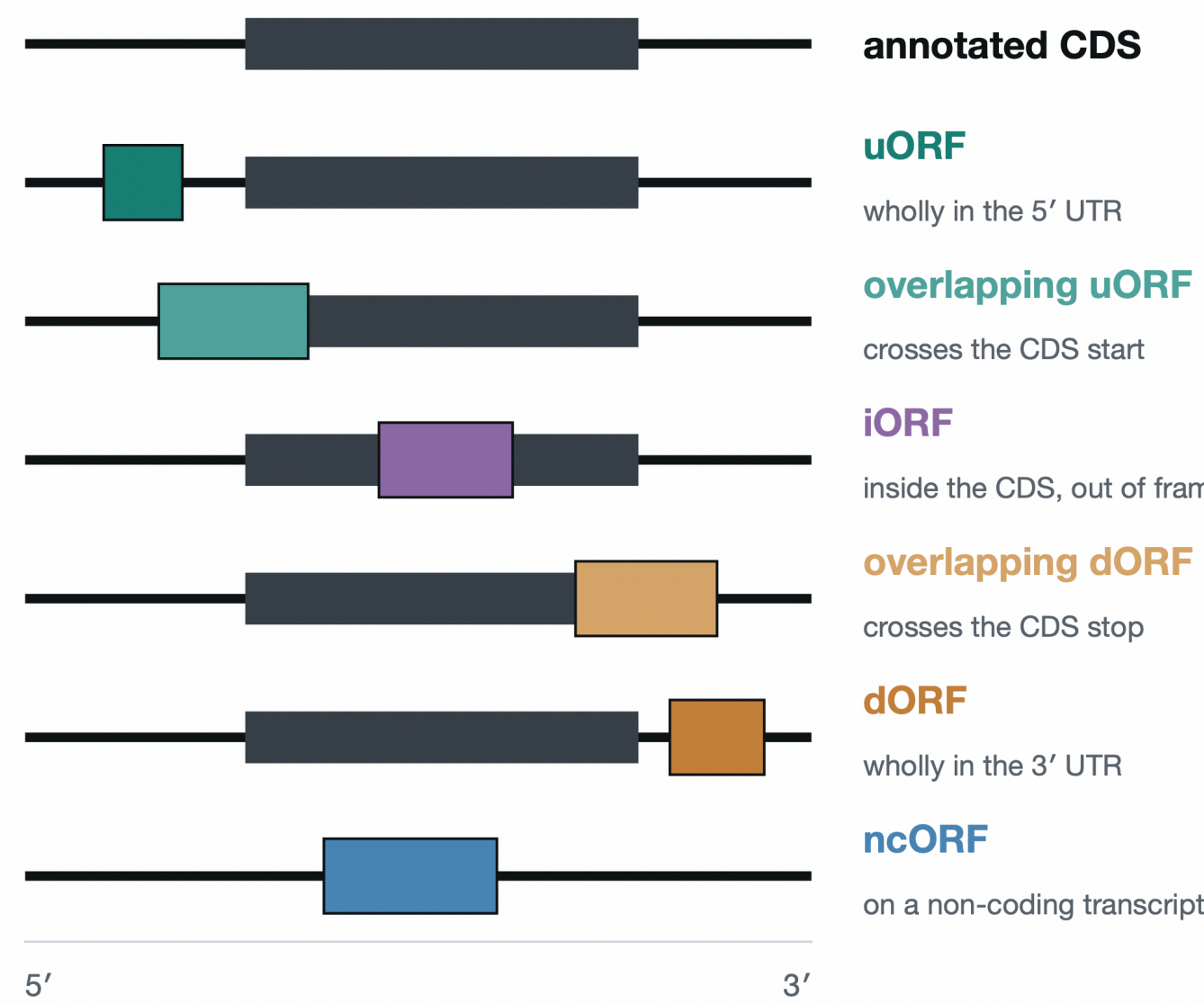
- Call translated ORFs in samples that were never ribosome-profiled
- Build proteogenomic search databases from RNA-seq alone
- Evaluate the model's ability to replace experimental Ribo-seq data

1. RIBO-SEQ AND NOVEL ORFs

A Ribosome profiling (Ribo-seq)



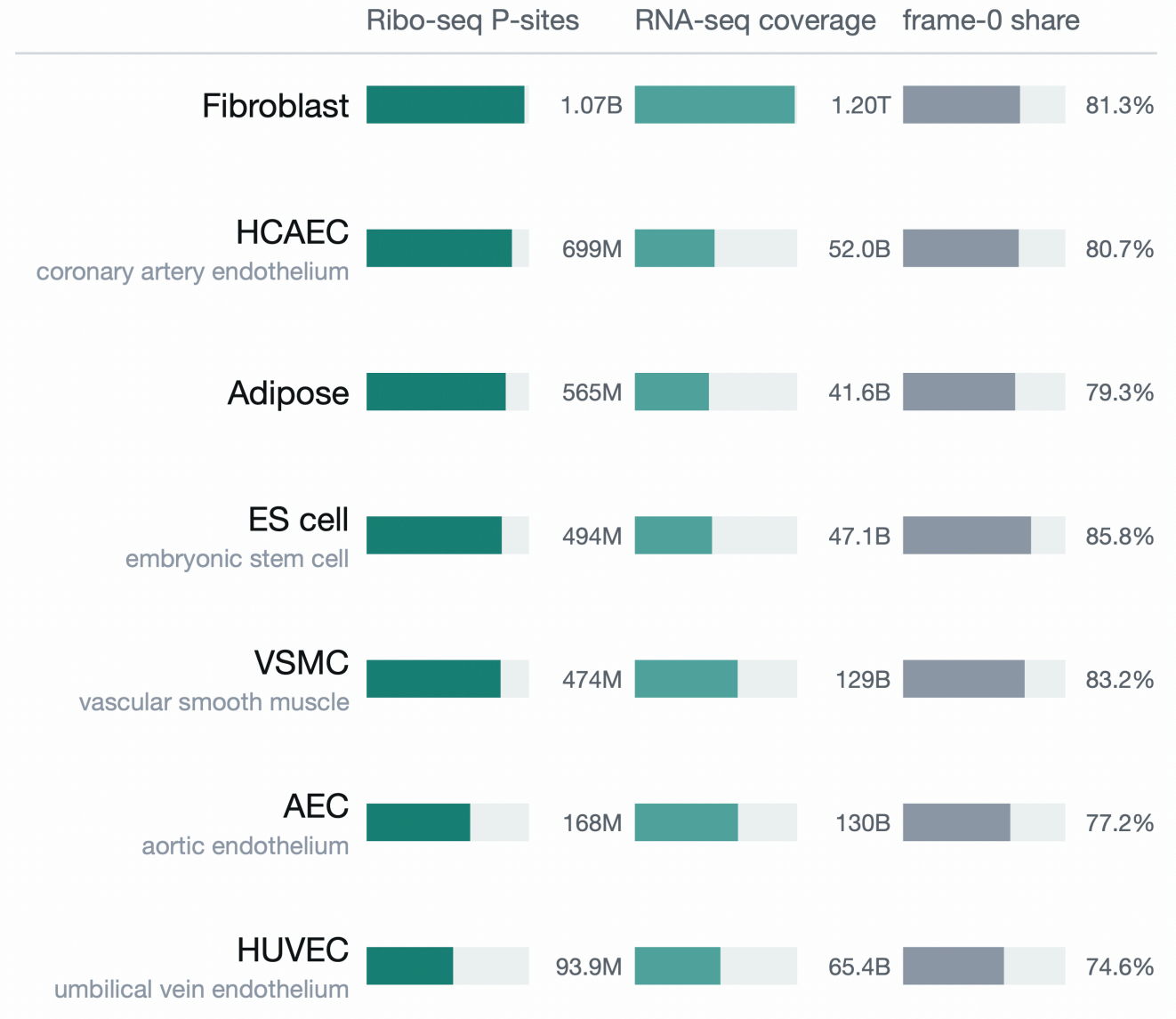
B Species of open reading frame



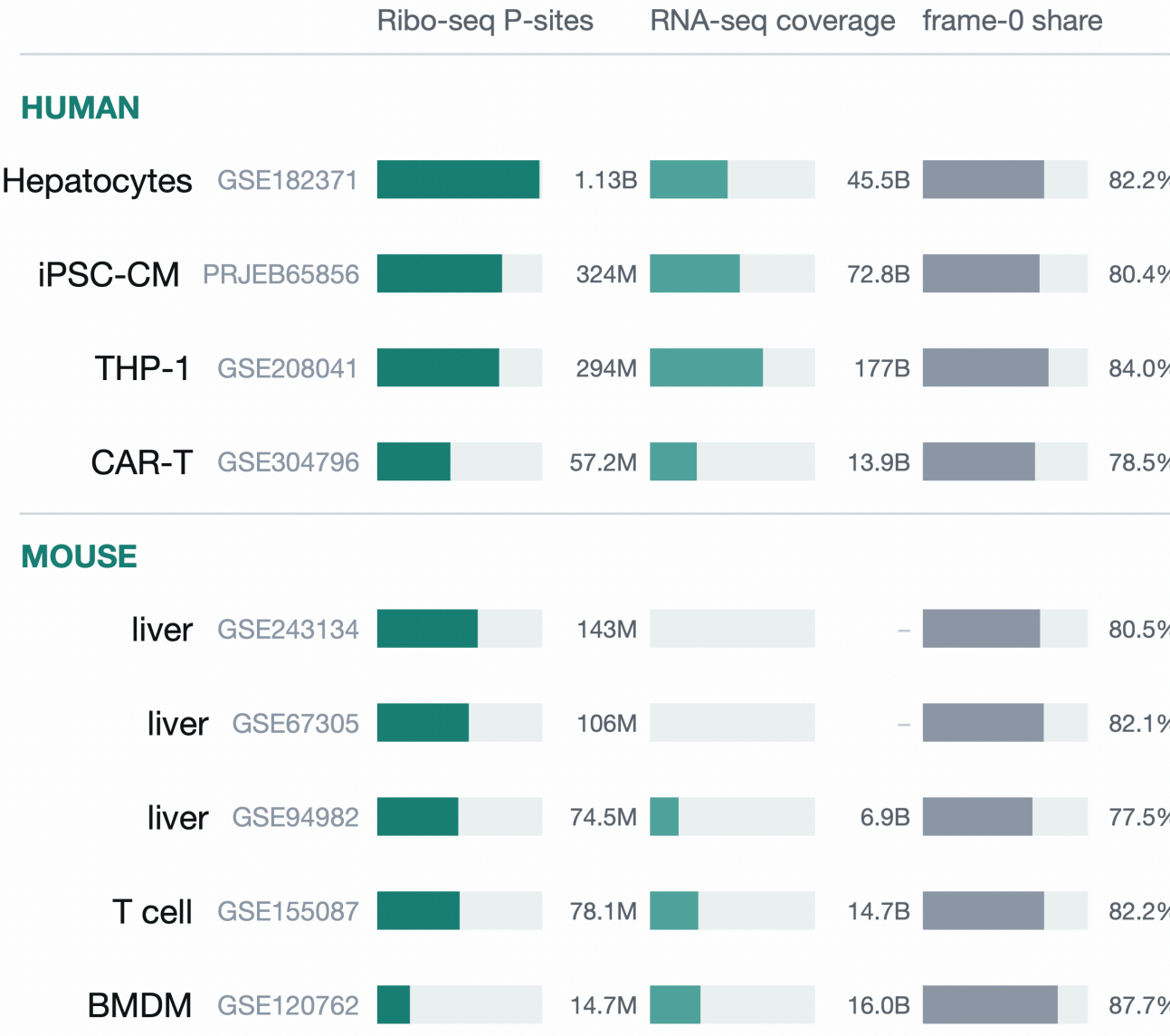
(A) Ribosome profiling digests unprotected RNA, sequences the ~30 nt fragments an elongating ribosome shields, and assigns each one to the codon in the ribosome's P site. Translation therefore reads out as P-site counts that step three nucleotides at a time. The trace is real observed data, not a schematic: CCL2 in human iPSC-derived cardiomyocytes (PRJEB65856), whose 100-codon ORF holds 99.0% of the transcript's P-sites, with 85.2% of those in frame 0. (B) The seven species of open reading frame this project calls, drawn 5' to 3'. Only the annotated CDS is in the reference; the other six are what the model is being asked to find.

2. TRAINING AND VALIDATION DATASETS

A Trained on



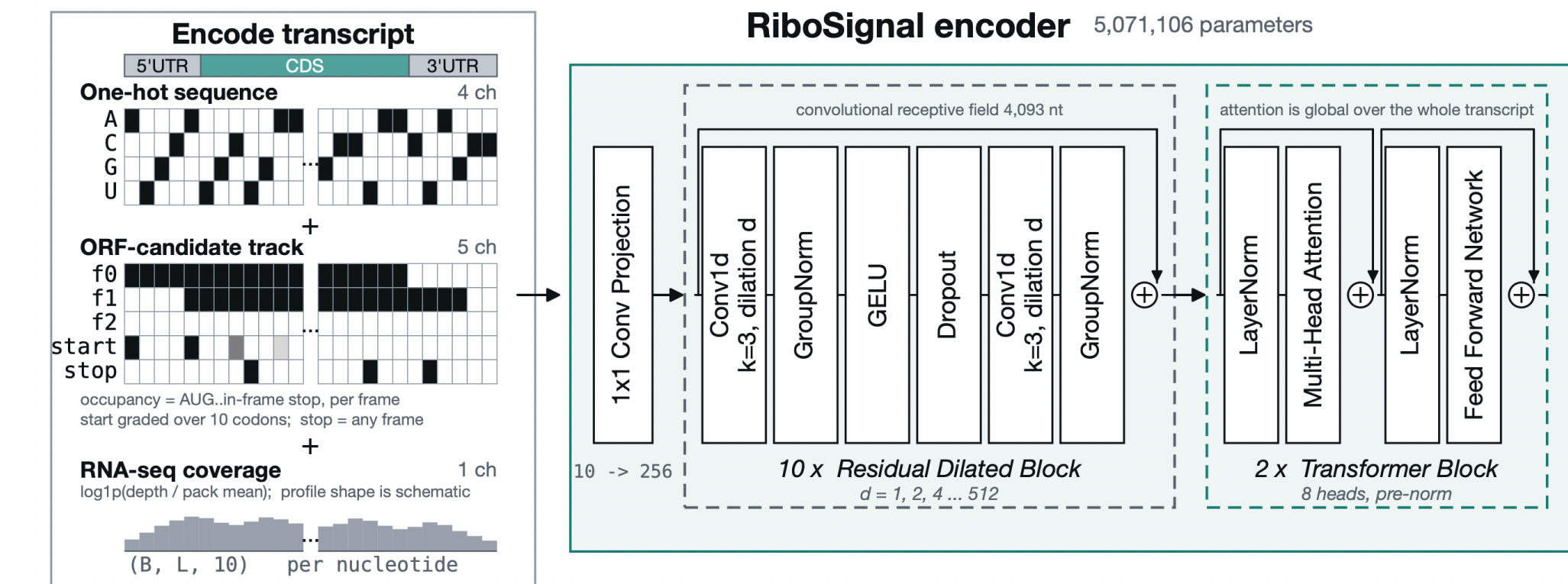
B Held out



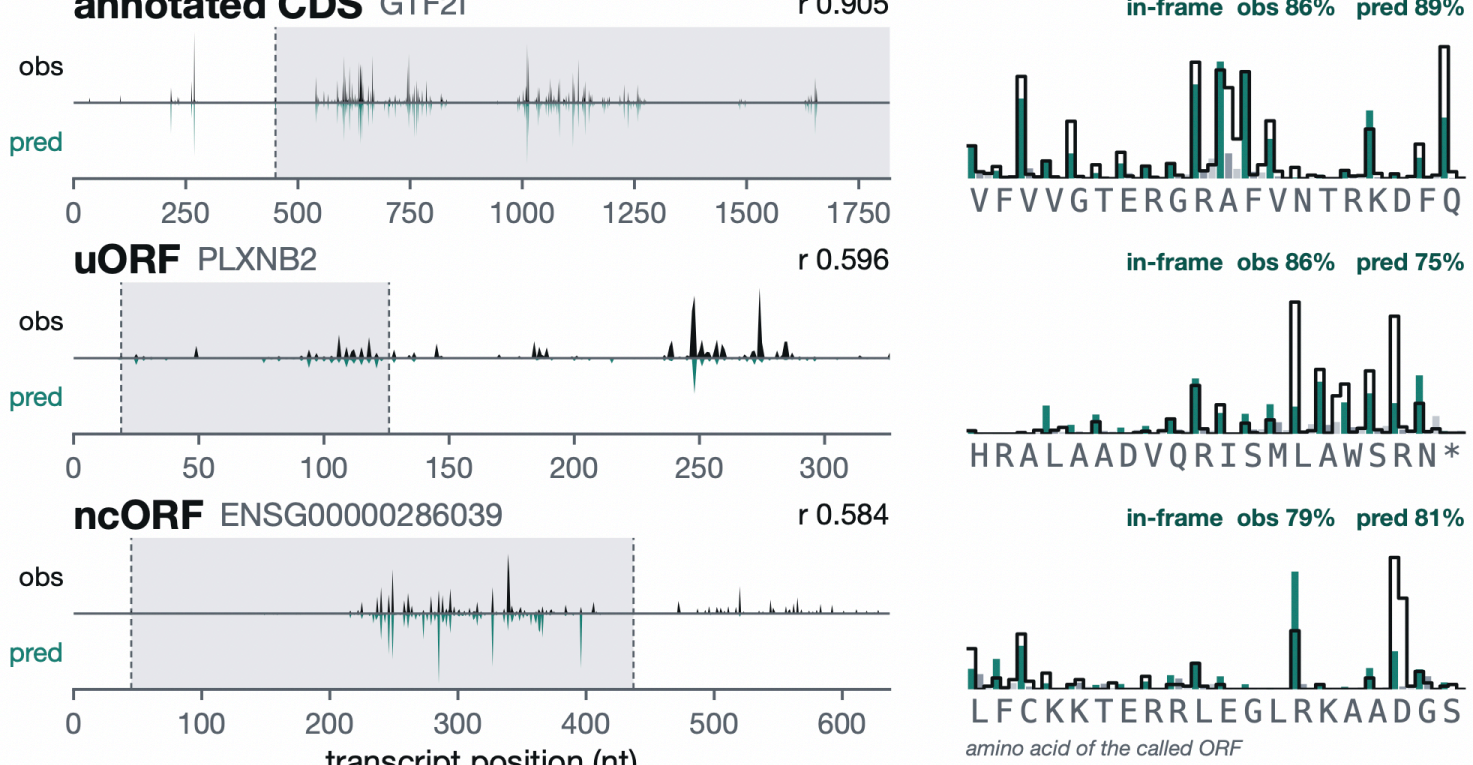
Seven human cell types from one study (GSE182371), 69 Ribo-seq libraries; brain, the study's ninth, was dropped before training on low periodicity. The three mouse livers come from three independent studies and are compared in panel 4a. P-sites count footprints while RNA coverage sums per-nucleotide read depth, so the two columns sit on separate log axes and are not comparable to each other. Frame-0 share runs from the 33% no-periodicity floor to 100%.

3. MODEL AND PREDICTIONS

A model architecture



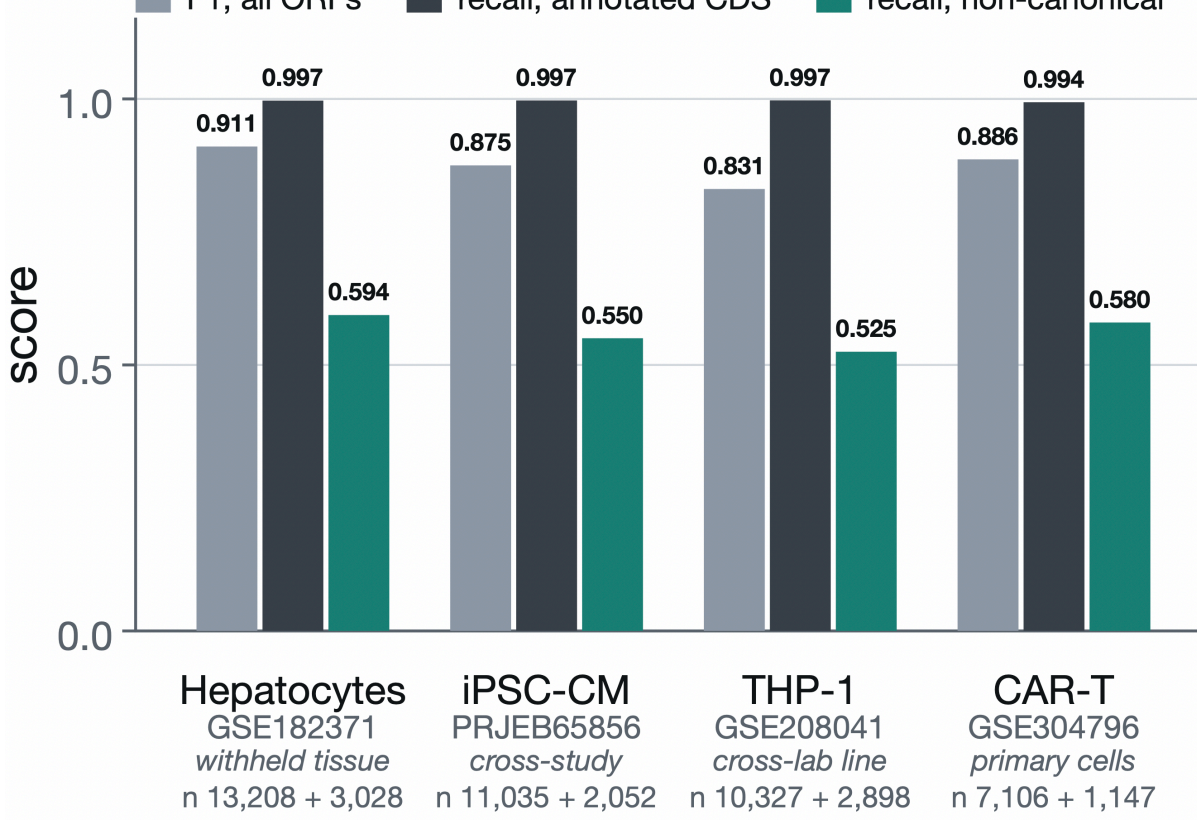
B observed vs predicted



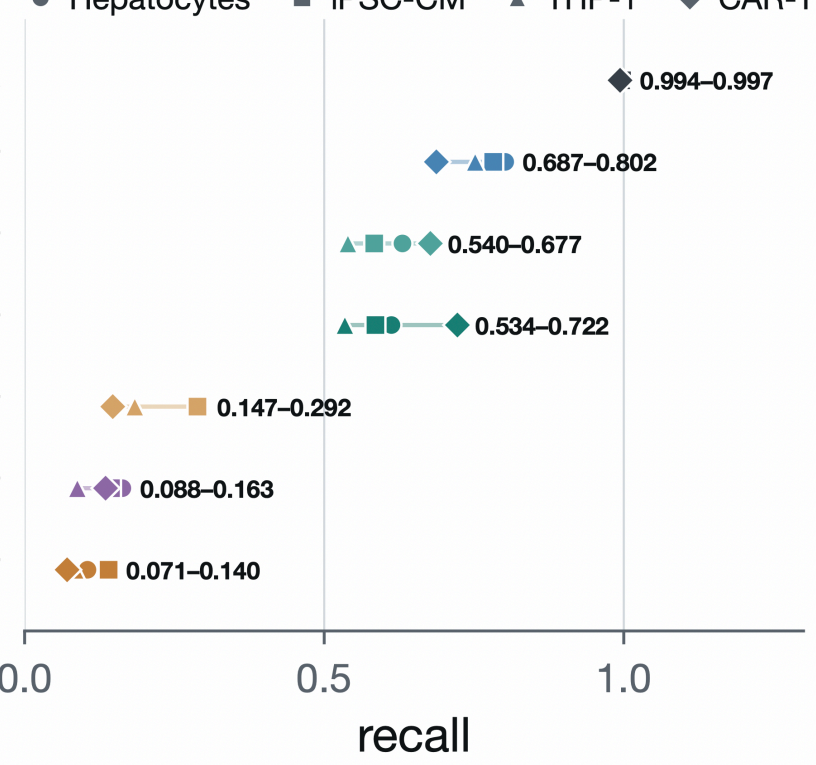
(A) Ten residual dilated blocks give a 4,093 nt receptive field, then two transformer blocks read the whole transcript; 5,071,106 parameters. The profile head is scale-free, so sequencing depth is isolated in the count head. No Ribo-seq is used at inference: the inputs are sequence and RNA-seq coverage only. The ORF-candidate track encodes every candidate ORF, not one, and the coverage profile drawn in the input block is schematic. (B) Held-out hepatocytes, one exemplar per ORF class, curated rather than randomly sampled. The annotated CDS row is cropped: its called ORF runs to 3,323 nt but the window ends at 1,819, where the observed signal stops. The correlation 'r' is the whole-transcript Pearson.

4. HELD-OUT HUMAN VALIDATION

A held-out human ORF calling



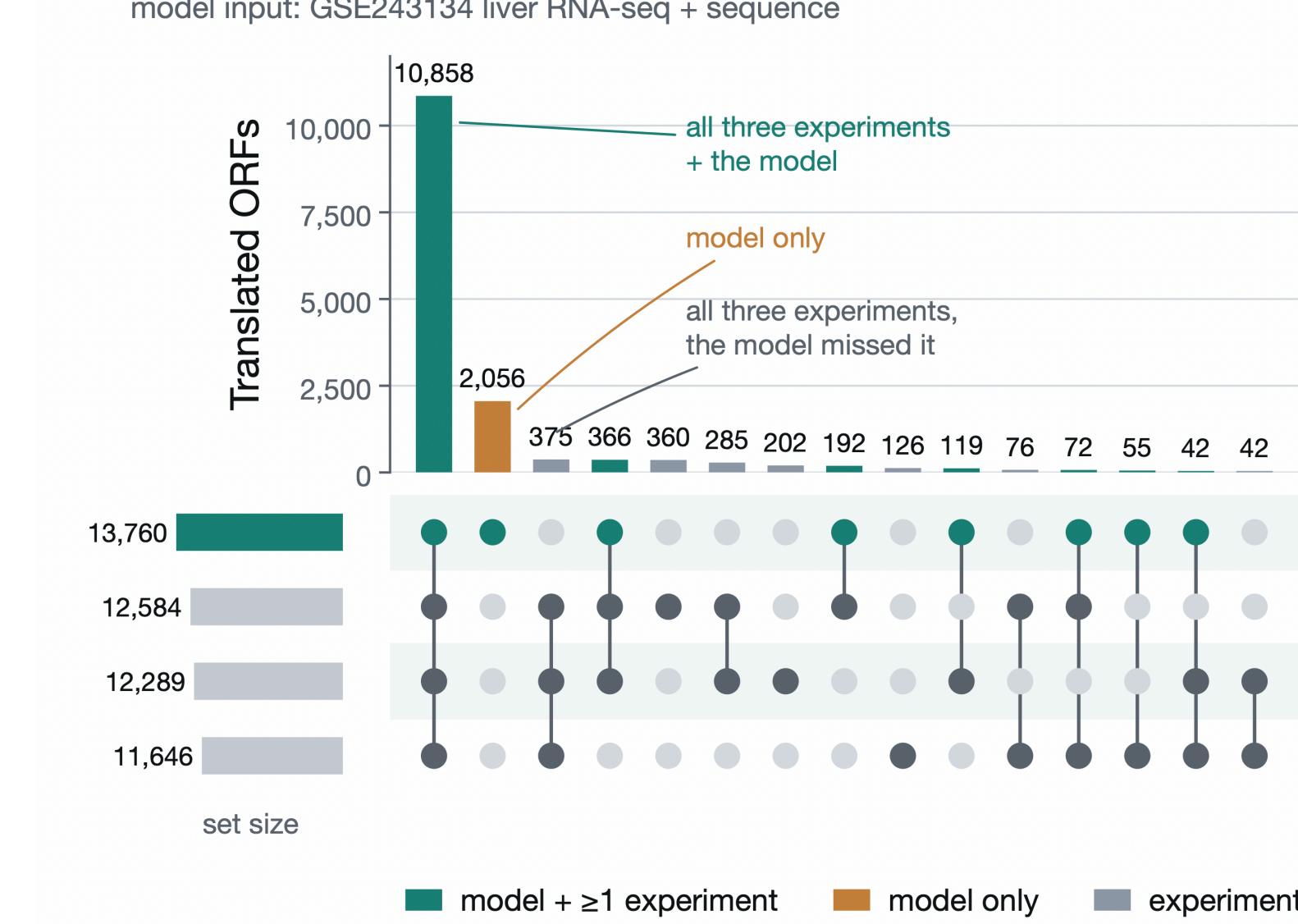
B recall by ORF class



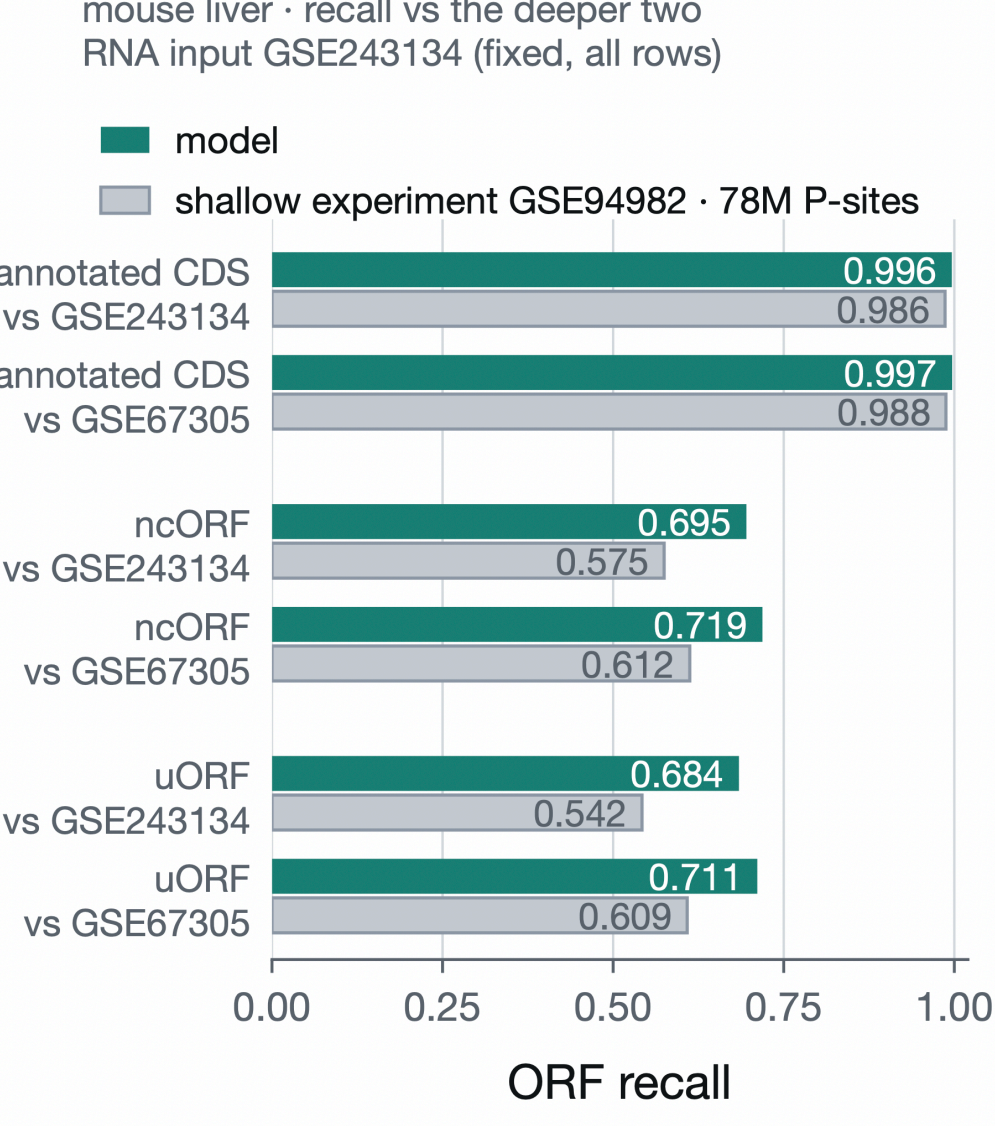
Standalone: RNA-seq and sequence only, no Ribo-seq for the query, scored against each dataset's own RiboCode calls. n in (A) = that dataset's reference transcript space; class values in (B) are RECALL, not F1, because this project defines precision over all ORFs only.

5. MOUSE LIVER: CROSS-STUDY COMPARISON

A Model agreement with three mouse liver Ribo-seq datasets



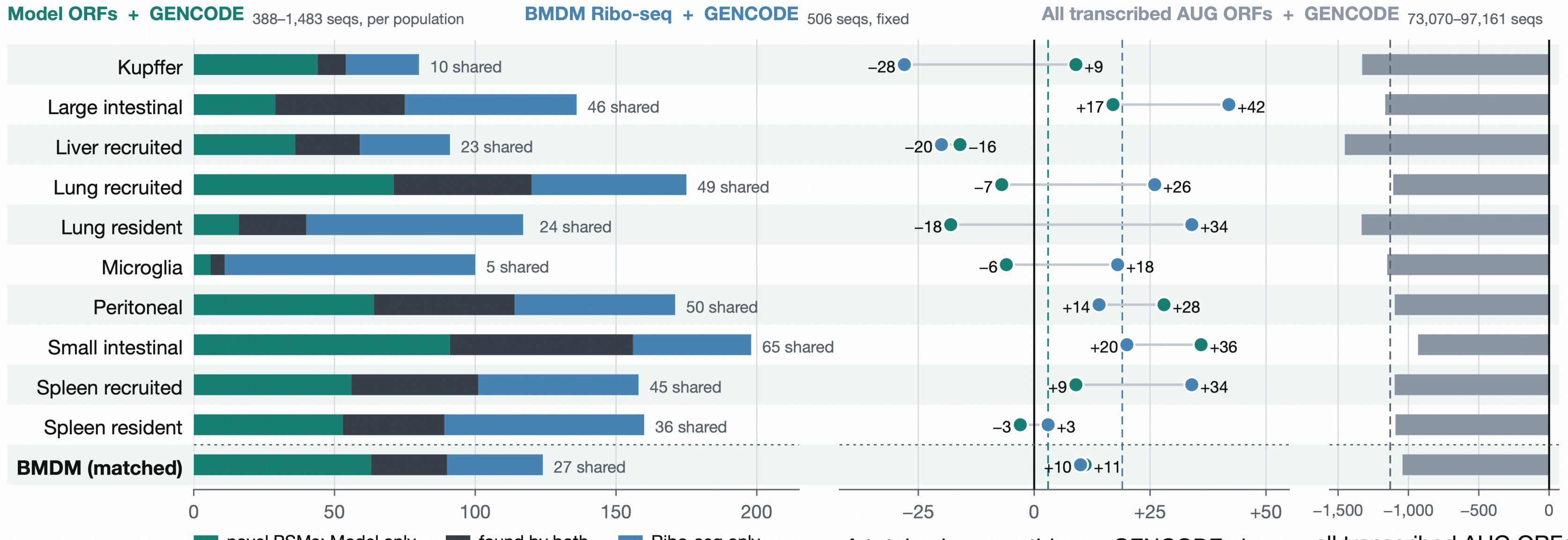
B Model vs a shallow experiment



(A) Where the standalone model agrees and disagrees with three independent mouse-liver Ribo-seq experiments, across all ORF classes; an F1 cannot show this. (B) Recall of the model against the recall of the shallowest of the three experiments, both scored on the two deeper ones. The model does not beat deep Ribo-seq: it loses to both deeper experiments on recall, and to all three on precision and therefore F1. The win is bounded to a shallow experiment, at no experimental cost.

6. PROTEOGENOMIC APPLICATION

Proteogenomics in primary mouse macrophages: three ORF databases, one set of mass spectra



Mouse macrophage populations (PRJNA482293), tryptic, 30 aa ORF floor, 18 mZML per population so acquisition depth cannot confound the comparison. Model ORFs are predicted from RNA-seq and sequence alone, with no Ribo-seq at inference. Novel PSMs are counted at each arm's own 1% class-specific FDR. RAW 264.7 is excluded as a transformed cell line rather than primary tissue, and BMDM is excluded from the medians the dashed lines draw. Note the naive AUG null on its own scale: a database of every transcribed AUG ORF loses about a thousand peptides per population to search-space inflation.

CONCLUSIONS, FUNDING AND REFERENCES

KEY TAKEAWAYS

- Standalone ORF calling reaches F1 0.831 to 0.911 on four independent human held-outs, with no Ribo-seq for the query (Figure 4)
- Recall is not uniform: 0.996 on annotated CDS against 0.098 on dORF (Figure 4)
- In a comparison of ORF calling in mouse liver samples, the model had better recall than a real, high quality riboseq experiment (Figure 5)
- Model predictions can be used to build a compact proteomics database for novel PSM discovery without inflating decoy databases (Figure 6)

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